Haotian Tang

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EDUCATION

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) | Abu Dhabi, UAE — Aug. 2026 – May 2030 (Expected)
PhD in Computational Biology
Supervisors: Eduardo Beltrame, PhD; Jun Wen, PhD

Columbia University | New York, NY, USA — Sep. 2023 – May 2025
Master of Science in Biostatistics | Statistical Genetics | GPA: 3.94/4.00

University of Toronto | Toronto, Canada — Sep. 2018 – Jun. 2022
Honours Bachelor of Science | Specialist: Pharmacology; Major: Genome Biology; Minor: Mathematics

TECHNICAL SKILLS

- **Programming & Computing:** Python, R, Bash, Linux, Git, SLURM/HPC; NumPy, pandas, SciPy, scikit-learn.
- **Machine Learning & Statistical Modeling:** PyTorch, graph neural networks, hypergraph learning, representation learning, tensor decomposition, spectral clustering, Gaussian mixture models, simulation, multivariable regression.
- **Single-cell Genomics:** Scanpy, AnnData, scVI, scHPL, pseudobulk analysis, PyDESeq2, Memento, scRNA-seq cell-type annotation and differential expression.
- **Genomics & Bioinformatics:** WGS/VCF processing, bcftools, bedtools, samtools, PLINK, GATK, variant QC/harmonization, rare-variant analysis, genomic interval operations.

SELECTED RESEARCH EXPERIENCE

PhD Researcher, MBZUAI | Abu Dhabi, United Arab Emirates — Sep. 2026 – Present
Supervisor: Alla Mikheenko, PhD
Research Topic: Rare-variant analysis of TDP-43-associated splicing dysregulation in amyotrophic lateral sclerosis

- Developed an HPC-based genomic analysis pipeline integrating whole-genome sequencing with TDP-43 binding regions, including QC and harmonization of more than 3,400 WGS VCFs, variant catalog construction, and functional annotation with gnomAD, VEP, and SpliceAI.
- Performed rare-variant burden and robustness analyses using target-versus-local-control comparisons, project-stratified models, Firth-regression sensitivity analyses, ancestry checks, and variant- and gene-level influence diagnostics.
- Built a pilot RNA-seq splice-junction validation framework for prioritized variants, comparing junction usage and read-level evidence across candidate loci to evaluate predicted splicing disruptions.

PhD Researcher, MBZUAI | Abu Dhabi, United Arab Emirates — Aug. 2026 – Present
Supervisor: Eduardo Beltrame, PhD
Research Topic: Single-cell transcriptomic analysis of population-specific immune responses to *Plasmodium falciparum* malaria

- Reanalyzed PBMC single-cell RNA-seq data from 126 Fulani and Mossi children in Burkina Faso, comprising 88,784 post-QC cells, to investigate population- and infection-associated immune variation.
- Built a reproducible single-cell annotation pipeline using scVI latent representations and donor-balanced hierarchical classification with scHPL, resolving major T-cell, B-cell, NK-cell, and monocyte populations.
- Constructed donor-by-cell-type pseudobulk profiles from raw UMI counts and performed differential expression analyses with PyDESeq2 and Memento, including sensitivity and permutation-based calibration analyses to evaluate robustness of population- and infection-associated signals.

Research Collaborator, Computational Pharmacogenomics | Remote — Jul. 2025 – Present
Supervisor: Jun Wen, PhD
Research Topic: HyperADRs: hypergraph learning for drug–gene–adverse drug reaction prediction

- Designed a biologically informed ADR categorization framework that reduced label sparsity and improved AUPR and mean reciprocal rank on PharmGKB drug–gene–ADR prediction tasks.
- Developed a relational hypergraph learning framework over drug–gene–ADR triplets using query-conditioned Hypergraph Convolutional Networks and benchmarked it against MLP, GCN, GAT, and HGNN baselines under inductive splits.
- Integrated causal gene/target, pathway, and protein–protein interaction evidence from PharmGKB and DrugBank to improve biological interpretability; co-led experimental design, model evaluation, figures, reproducibility, and manuscript preparation for the resulting co-first-author preprint.

SELECTED PUBLICATIONS

- Cai, Z.*, **Tang, H.***, Gao, S., Zhou, B., Zhao, J., & Wen, J. HyperADRs: A hierarchical hypergraph framework for drug–gene–ADR prediction. *arXiv:2512.00137*, 2025. ***Equal contribution.** arXiv:2512.00137.
- Hark, L. A., Gorroochurn, P., **Tang, H.**, et al. (2025). Improvement in vision-related quality-of-life using the NEI-VFQ-9 over 1-year in the Manhattan Vision Screening and Follow-up Study (NYC-SIGHT). *Graefe's Archive for Clinical and Experimental Ophthalmology*, 263(7), 2069–2079. doi:10.1007/s00417-024-06727-z.
- Morse, A. R., Hark, L. A., Seiple, W. H., Gorroochurn, P., **Tang, H.**, et al. (2025). Association of behavioral factors with activation in patients with age-related macular degeneration or diabetic retinopathy. *Clinical Ophthalmology*, 19, 3059–3069. doi:10.2147/OPTH.S542352.
- Morse, A. R., Hark, L. A., Seiple, W., Gorroochurn, P., **Tang, H.**, et al. (2025). Modifying locus of control beliefs may improve activation in patients with age-related macular degeneration or diabetic retinopathy. *Investigative Ophthalmology & Visual Science*, 66(8), 699.

ADDITIONAL RESEARCH EXPERIENCE

Research Assistant, Department of Biostatistics, Columbia University | New York, NY, USA Apr. 2024 – May 2025
Supervisor: Shuang Wang, PhD
Research Topic: Multi-omics integration and disease subtyping
- Developed simulation studies using spectral clustering and Gaussian mixture models to benchmark multi-omics integration strategies and tune clustering parameters after reviewing more than a decade of methodological literature.

Research Assistant, Department of Biostatistics, Columbia University | New York, NY, USA Sep. 2024 – May 2025
Supervisor: Tian Gu, PhD
Research Topic: Cancer analysis using the All of Us Research Program
- Applied angle-based transfer learning (AngleTL) to large-scale breast and prostate cancer data from the All of Us Research Program, incorporating EHR-derived cancer phenotypes and clinical variables.

Research Assistant, Departments of Biostatistics and Ophthalmology, Columbia University | New York, NY, USA Feb. 2024 – Aug. 2024
Supervisor: Prakash Gorroochurn, PhD; Lisa Hark, PhD, MBA
Research Topic: Longitudinal analysis of vision-related quality of life in NYC-SIGHT
- Performed longitudinal and multivariable statistical analyses of randomized clinical trial data to identify predictors of improvement in NEI-VFQ-9 vision-related quality-of-life scores; contributed to a peer-reviewed publication.

Research Assistant, Department of Ophthalmology, Columbia University | New York, NY, USA Apr. 2024 – Aug. 2024
Supervisor: Alan Morse, PhD
Research Topic: Clinical and demographic analysis in age-related macular degeneration and diabetic retinopathy
- Compared demographic and clinical characteristics across AMD and diabetic retinopathy subgroups and contributed analyses to subsequent peer-reviewed publications.

Research Student, Department of Pharmacology, University of Toronto | Toronto, Canada Sep. 2021 – Apr. 2022
Supervisor: Walter Swardfager, PhD
Research Topic: Genetic analysis of CYP450 and soluble epoxide hydrolase pathways in Alzheimer's disease
- Analyzed ADNI genetic, biomarker, and cognitive phenotype data to evaluate SNPs related to lipid metabolism across cognitively normal, mild cognitive impairment, and dementia groups.

TEACHING EXPERIENCE

Teacher and Lecturer, Shanghai United International School Hefei Campus | Hefei, China Aug. 2025 – Jun. 2026
- Instructed International Advanced Level Mathematics and Further Mathematics, including lectures, assessment, and grading.

Teaching Assistant, Columbia University | New York, NY, USA Sep. 2024 – Dec. 2024
- Supported graduate Probability and Fundamentals of Toxicology through assignment development, grading, instruction, and student support.